```
(*This notebook computes the Euler factors of the U^{2}-sum of the paper*)

(*Set x=1/p^u[1],y=1/p^s and;real parts of u[1] and s are 1/2+and 1+*)

In[1]:= SetDirectory[NotebookDirectory[]]; << gln.m

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SetDelayed: Tag UpperTriangularMatrixQ in UpperTriangularMatrixQ[m_] is Protected.

GL(n)PACK loaded.

In[2]:= Schu[n_, d_, nota_ : a] :=
         SchurPolynomial[Table[nota[i], {i, 1, d}], Join[ConstantArray[0, d - 2], {n}]]

In[3]:= InvL[X_, d_, nota_ : a] := Product[(1 - nota[i] X), {i, 1, d}]

In[4]:= InvRSL[X_, d_, nota1_ : a, nota2_ : b] :=
         Product[Product[(1 - nota1[i] * nota2[j] X), {i, 1, d}], {j, 1, d}]

In[5]:= (*PGL2*)

In[6]:= (*Computing L_{p}(u[1];s,Pi[1],Pi[2])*)

In[7]:= L2[x_, y_] := Sum[Schu[r, 2, a] x ^ r, {r, 1, Infinity}] +
          Sum[Schu[n, 2, b] y ^ n * Sum[Schu[n + r, 2, a] x ^ r, {r, 1, Infinity}], {n, 1, Infinity}]

In[8]:= P2[x_, y_] := Cancel[InvL[x, 2, a] * InvRSL[y, 2, a, b] * L2[x, y]]

In[9]:= P2[x, y]
```

Out[9]= $x\, a[1] + x\, a[2] - x^2\, a[1] \times a[2] - x\, y\, a[1] \times a[2] \times b[1] - x\, y\, a[1] \times a[2] \times b[2] + x^2\, y^2\, a[1]^2\, a[2]^2\, b[1] \times b[2]$

```
In[10]:= P20[x_, y_] := Assuming[a[1] × a[2] == b[1] × b[2] == 1, FullSimplify[P2[x, y]]]

In[11]:= P20[x, y]
```

Out[11]= $x\left(x\left(-1 + y^2\right) + a[1] + a[2] - y\,(b[1] + b[2])\right)$

```
In[12]:= M20 = MonomialList[P20[x, y], {x, y, a[1], a[2], b[1], b[2]}];
         sorted = SortBy[M20, Total[Exponent[#, {x, y, a[1], a[2], b[1], b[2]}]] &];
         Total[sorted]
```

Out[14]= $-x^2 + x^2\, y^2 + x\, a[1] + x\, a[2] - x\, y\, b[1] - x\, y\, b[2]$

```
In[15]:= (*The following outputs non-negligible terms in the Euler product*)

In[16]:= Select[M20, Exponent[#, x] / 2 + Exponent[#, y] -
            (Exponent[#, a[1]] + Exponent[#, a[2]] + Exponent[#, b[1]] + Exponent[#, b[2]]) * 7 / 64 ≤ 1.00001 &]
```

Out[16]= $\left\{-x^2,\ x\, a[1],\ x\, a[2]\right\}$

```
In[17]:= (*Considering also the factor (1-1/p^(2-w));Re(2-w)=1-\epsilon*)

In[18]:= Select[M20, 0.9999 + Exponent[#, x] / 2 + Exponent[#, y] -
           (Exponent[#, a[1]] + Exponent[#, a[2]] + Exponent[#, b[1]] + Exponent[#, b[2]]) * 0.49 ≤ 1.00001 &]
Out[18]=
         {}

In[19]:= (*PGL3*)

In[20]:= (*Computing L_{p}(u[1];s,Pi[1],Pi[2])---I*)

In[21]:= L3Pol1[x_, y_] :=
          Cancel[InvRSL[y, 3, a, b] * x * Sum[Schu[n, 3, b] × Schu[n + 1, 3, a] y ^ n, {n, 1, Infinity}]]

In[22]:= Mono1 = MonomialList[L3Pol1[x, y], {x, y, a[1], a[2], a[3], b[1], b[2], b[3]}];

In[23]:= M1u = Assuming[a[1] × a[2] × a[3] == 1 && b[1] × b[2] × b[3] == 1, FullSimplify[Mono1]];

In[24]:= (*Checking:fully simplified already?*)

In[25]:= Select[M1u, And @@ (Not[FreeQ[#, #2]] & @@@ Thread[{#, {a[1], a[2], a[3]}}]) &]
Out[25]=
         {}

In[26]:= Select[M1u, And @@ (Not[FreeQ[#, #2]] & @@@ Thread[{#, {b[1], b[2], b[3]}}]) &]
Out[26]=
         {}

In[27]:= NP1 = Total[Select[M1u,
           0.000001 + Exponent[#, x] / 2 + Exponent[#, y] - (Exponent[#, a[1]] + Exponent[#, a[2]] + Exponent[
                  #, a[3]] + Exponent[#, b[1]] + Exponent[#, b[2]] + Exponent[#, b[3]]) * (5 / 14) ≤ 1 &]];

In[28]:= (*Coeff of xy of NP1*)

In[29]:= SymmetricReduction[SymmetricReduction[Coefficient[Collect[Cancel[NP1 / x], y], y, 1],
           {b[1], b[2], b[3]}, {B[1], B[2], 1}][[1]], {a[1], a[2], a[3]}, {A[1], A[2], 1}][[1]]
Out[29]=
```

$A[1]^2 B[1] - A[2] \times B[1]$

```
In[30]:= (*Coeff of xy^2 of NP1*)

In[31]:= SymmetricReduction[SymmetricReduction[Coefficient[Collect[Cancel[NP1 / x], y], y, 2],
           {b[1], b[2], b[3]}, {B[1], B[2], 1}][[1]], {a[1], a[2], a[3]}, {A[1], A[2], 1}][[1]]
Out[31]=
```

$3 \left(B[1]^2 - 2 B[2]\right) + 6 B[2] - A[1]^3 B[2] + A[1] \times A[2] \left(-B[1]^2 + 2 B[2]\right)$

```
In[32]:= Collect[3 (B[1]^2 - 2 B[2]) + 6 B[2] - A[1]^3 B[2] + A[1] × A[2] (-B[1]^2 + 2 B[2]), B[2]]
Out[32]=
```

$3 B[1]^2 - A[1] \times A[2] B[1]^2 + \left(-A[1]^3 + 2 A[1] \times A[2]\right) B[2]$

```
In[33]:= (*Computing L_{p}(u[1];s,Pi[1],Pi[2])---II*)
```

In[34]:=
```
L3Pol2[x_, y_] :=
  Cancel[InvRSL[y, 3, a, b] * x^2 * Sum[Schu[n, 3, b] × Schu[n - 1, 3, a] y^n, {n, 1, Infinity}]]
```

In[35]:=
```
Mono2 = MonomialList[L3Pol2[x, y], {x, y, a[1], a[2], a[3], b[1], b[2], b[3]}];
```

In[36]:=
```
M2u = Assuming[a[1] × a[2] × a[3] == 1 && b[1] × b[2] × b[3] == 1, FullSimplify[Mono2]];
```

In[37]:=
```
(*Checking:fully simplified already?*)
```

In[38]:=
```
Select[M2u, And @@ (Not[FreeQ[#, #2]] & @@@ Thread[{#, {a[1], a[2], a[3]}}]) &]
```

Out[38]=
```
{}
```

In[39]:=
```
Select[M2u, And @@ (Not[FreeQ[#, #2]] & @@@ Thread[{#, {b[1], b[2], b[3]}}]) &]
```

Out[39]=
```
{}
```

In[40]:=
```
NP2 = Total[
  Select[M2u, 0.000001 + Exponent[#, x] / 2 + Exponent[#, y] - (Exponent[#, a[1]] + Exponent[#, a[2]] +
        Exponent[#, a[3]] + Exponent[#, b[1]] + Exponent[#, b[2]] + Exponent[#, b[3]]) * (5 / 14) ≤ 1 &]]
```

Out[40]=
```
0
```